

Ex. No: 1

Print a 2D Array

PROGRAM:

```
import java.util.Scanner;

public class array21 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

            }

        }

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                System.out.print(matrix[i][j] + " ");

            }

            System.out.println();

        }

        sc.close();

    }

}
```

Ex. No: 2**Sum of All Elements****PROGRAM:**

```
import java.util.Scanner;

public class array22 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        int sum = 0;

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

                sum += matrix[i][j];

            }

        }

        System.out.println(sum);

        sc.close();

    }

}
```

Ex. No: 3**Largest Element****PROGRAM:**

```
import java.util.Scanner;

public class array23 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

            }

        }

        int largest = matrix[0][0];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                if (matrix[i][j] > largest) {

                    largest = matrix[i][j];

                }

            }

        }

        System.out.println(largest);

        sc.close();

    }

}
```

Ex. No: 4**Smallest Element****PROGRAM:**

```
import java.util.Scanner;

public class array24 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

            }

        }

        int smallest = matrix[0][0];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                if (matrix[i][j] < smallest) {

                    smallest = matrix[i][j];

                }

            }

        }

        System.out.println(smallest);

        sc.close();

    }

}
```

Ex. No: 5**Sum of Each Row****PROGRAM:**

```
import java.util.Scanner;

public class array25 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

            }

        }

        for (int i = 0; i < rows; i++) {

            int sum = 0;

            for (int j = 0; j < cols; j++) {

                sum += matrix[i][j];

            }

            System.out.println("Row " + (i + 1) + " Sum = " + sum);

        }

        sc.close();

    }

}
```

Ex. No: 6**Sum of Each Column****PROGRAM:**

```
import java.util.Scanner;

public class array26 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

            }

        }

        for (int j = 0; j < cols; j++) {

            int sum = 0;

            for (int i = 0; i < rows; i++) {

                sum += matrix[i][j];

            }

            System.out.println("Column " + (j + 1) + " Sum = " + sum);

        }

        sc.close();

    }

}
```

Ex. No: 7

Count Even and Odd Numbers

PROGRAM:

```
import java.util.Scanner;

public class array27 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        int even = 0;

        int odd = 0;

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

                if (matrix[i][j] % 2 == 0) {

                    even++;

                } else {

                    odd++;

                }

            }

        }

        System.out.println("Even = " + even);

        System.out.println("Odd = " + odd);

        sc.close();

    }

}
```

Ex. No: 8

Search an Element

PROGRAM:

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix[i][j] = sc.nextInt();

            }

        }

        int key = sc.nextInt();

        boolean found = false;

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                if (matrix[i][j] == key) {

                    System.out.println("Found at Row " + (i + 1) + " Column " + (j +
1));

                    found = true;

                    break;

                }

            }

        }

        if (found) {
```



```
        break;
    }
}
if (!found) {
    System.out.println("Not Found");
}
sc.close();
}
}
```

Ex. No: 9 **Transpose of a Matrix**

PROGRAM:

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();
        int cols = sc.nextInt();

        int[][] matrix = new int[rows][cols];

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        for (int j = 0; j < cols; j++) {
            for (int i = 0; i < rows; i++) {
```

```
        System.out.print(matrix[i][j] + " ");  
    }  
    System.out.println();  
}  
sc.close();  
}  
}
```

Ex. No: 10 Primary and Secondary Diagonal Sum

PROGRAM:

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int rows = sc.nextInt();  
        int cols = sc.nextInt();  
        int[][] matrix = new int[rows][cols];  
        for (int i = 0; i < rows; i++) {  
            for (int j = 0; j < cols; j++) {  
                matrix[i][j] = sc.nextInt();  
            }  
        }  
        int primarySum = 0;  
        int secondarySum = 0;  
        for (int i = 0; i < rows; i++) {  
            primarySum += matrix[i][i];  
            secondarySum += matrix[i][cols - 1 - i];  
        }  
        System.out.println("Primary Diagonal Sum: " + primarySum);  
        System.out.println("Secondary Diagonal Sum: " + secondarySum);  
    }  
}
```

```
        secondarySum += matrix[i][cols - 1 - i];
    }
    System.out.println("Primary Diagonal Sum = " + primarySum);
    System.out.println("Secondary Diagonal Sum = " + secondarySum);
    sc.close();
}
}
```

Ex. No: 11

Matrix Addition

PROGRAM:

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();
        int cols = sc.nextInt();

        int[][] matrix1 = new int[rows][cols];
        int[][] matrix2 = new int[rows][cols];
        int[][] sum = new int[rows][cols];

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                matrix1[i][j] = sc.nextInt();
            }
        }

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
```

```
        matrix2[i][j] = sc.nextInt();
    }
}
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        sum[i][j] = matrix1[i][j] + matrix2[i][j];
    }
}
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        System.out.print(sum[i][j] + " ");
    }
    System.out.println();
}

sc.close();
}
}
```

Ex. No: 12**Matrix Subtraction****PROGRAM:**

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] matrix1 = new int[rows][cols];

        int[][] matrix2 = new int[rows][cols];

        int[][] result = new int[rows][cols];

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix1[i][j] = sc.nextInt();

            }

        }

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                matrix2[i][j] = sc.nextInt();

            }

        }

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                result[i][j] = matrix1[i][j] - matrix2[i][j];

            }

        }

    }

}
```

```
    for (int i = 0; i < rows; i++) {  
        for (int j = 0; j < cols; j++) {  
            System.out.print(result[i][j] + " ");  
        }  
        System.out.println();  
    }  
    sc.close();  
}  
}
```

Ex. No: 13 **Matrix Multiplication**

PROGRAM:

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int r1 = sc.nextInt();  
        int c1 = sc.nextInt();  
        int[][] A = new int[r1][c1];  
        for (int i = 0; i < r1; i++)  
            for (int j = 0; j < c1; j++)  
                A[i][j] = sc.nextInt();  
  
        int r2 = sc.nextInt();  
        int c2 = sc.nextInt();  
  
        int[][] B = new int[r2][c2];
```

```
    for (int i = 0; i < r2; i++)
        for (int j = 0; j < c2; j++)
            B[i][j] = sc.nextInt();
    int[][] C = new int[r1][c2];
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c2; j++) {
            for (int k = 0; k < c1; k++) {
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c2; j++)
            System.out.print(C[i][j] + " ");
        System.out.println();
    }
}
```

Ex. No: 14**Reverse Each Row****PROGRAM:**

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] a = new int[rows][cols];

        for (int i = 0; i < rows; i++)

            for (int j = 0; j < cols; j++)

                a[i][j] = sc.nextInt();

        for (int i = 0; i < rows; i++) {

            for (int j = cols - 1; j >= 0; j--)

                System.out.print(a[i][j] + " ");

            System.out.println();

        }

    }

}
```


Ex. No: 15**Reverse Each Column****PROGRAM:**

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] a = new int[rows][cols];

        for (int i = 0; i < rows; i++)

            for (int j = 0; j < cols; j++)

                a[i][j] = sc.nextInt();

        for (int i = rows - 1; i >= 0; i--) {

            for (int j = 0; j < cols; j++)

                System.out.print(a[i][j] + " ");

            System.out.println();

        }

    }

}
```

Ex. No: 16**Print Boundary Elements****PROGRAM:**

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] a = new int[rows][cols];

        for (int i = 0; i < rows; i++)

            for (int j = 0; j < cols; j++)

                a[i][j] = sc.nextInt();

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                if (i == 0 || i == rows - 1 || j == 0 || j == cols - 1)

                    System.out.print(a[i][j] + " ");

                else

                    System.out.print(" ");

            }

            System.out.println();

        }

    }

}
```

Ex. No: 17

Rotate Matrix 90° Clockwise

PROGRAM:

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int m = sc.nextInt();

        int[][] a = new int[n][m];

        for (int i = 0; i < n; i++)

            for (int j = 0; j < m; j++)

                a[i][j] = sc.nextInt();

        for (int j = 0; j < m; j++) {

            for (int i = n - 1; i >= 0; i--)

                System.out.print(a[i][j] + " ");

            System.out.println();

        }

    }

}
```

Ex. No: 18**Spiral Traversal****PROGRAM:**

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] a = new int[rows][cols];

        for (int i = 0; i < rows; i++)

            for (int j = 0; j < cols; j++)

                a[i][j] = sc.nextInt();

        int top = 0, bottom = rows - 1;

        int left = 0, right = cols - 1;

        while (top <= bottom && left <= right) {

            for (int i = left; i <= right; i++)

                System.out.print(a[top][i] + " ");

            top++;

            for (int i = top; i <= bottom; i++)

                System.out.print(a[i][right] + " ");

            right--;

            if (top <= bottom) {

                for (int i = right; i >= left; i--)

                    System.out.print(a[bottom][i] + " ");

                bottom--;

            }

        }

    }

}
```

```

        if (left <= right) {
            for (int i = bottom; i >= top; i--)
                System.out.print(a[i][left] + " ");
            left++;
        }
    }
}

```

Ex. No: 19

Wave Traversal

PROGRAM:

```

import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();

        int cols = sc.nextInt();

        int[][] a = new int[rows][cols];

        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                a[i][j] = sc.nextInt();

        for (int j = 0; j < cols; j++) {

            if (j % 2 == 0) {

                for (int i = 0; i < rows; i++)

                    System.out.print(a[i][j] + " ");

            } else {

```

```

        for (int i = rows - 1; i >= 0; i--)
            System.out.print(a[i][j] + " ");
    }
}
}
}

```

Ex. No: 20

Zigzag Traversal

PROGRAM:

```

import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();
        int cols = sc.nextInt();
        int[][] a = new int[rows][cols];
        for (int i = 0; i < rows; i++)
            for (int j = 0; j < cols; j++)
                a[i][j] = sc.nextInt();
        for (int i = 0; i < rows; i++) {
            if (i % 2 == 0) {
                for (int j = 0; j < cols; j++)
                    System.out.print(a[i][j] + " ");
            } else {
                for (int j = cols - 1; j >= 0; j--)
                    System.out.print(a[i][j] + " ");
            }
        }
    }
}

```

```
    }  
    System.out.println();  
}  
}  
}
```